

Week 5 Assignment: Tabular Q-Learning

A tested model-free agent with controlled exploration and oracle comparison

CMPSC 480 - Instructor Solution Deck

Instructor material - do not distribute with the question deck

Solution overview

This instructor deck provides a complete matched solution and grading guidance.

Coverage: Week 5: TD learning, exploration, terminal targets, and empirical evaluation

Points

10 total

Expected time

6-8 hours

Submission

Repository, learning curves,
and report

Instructor verification checklist

- ▶ Use Python 3.11 and NumPy only unless the prompt explicitly permits another package.
- ▶ Preserve the published interfaces and make all tie-breaking deterministic.
- ▶ State assumptions, validate inputs, and handle empty, terminal, and unreachable cases.
- ▶ Include focused tests whose names explain the defect each test would expose.
- ▶ Report measured evidence; do not replace experimental results with unsupported claims.
- ▶ Seed every source of randomness used in reported experiments.

Solution 1.a - Agent and exploration contract (1 points)

Question focus. Define observe, select_action, update, and evaluate-mode behavior.

Answer. The agent receives transitions from the environment, stores Q values keyed by state-action, selects only legal actions, updates from observations, and uses epsilon zero during evaluation.

Solution 1.b - Agent and exploration contract (1 points)

Question focus. Specify epsilon-greedy selection and a bounded decay schedule.

Answer. With probability epsilon choose uniformly among legal actions; otherwise choose a stable argmax. Decay is applied at a documented cadence and clipped to epsilon_min.

Solution 2.a - Q-learning update (1 points)

Question focus. State the update and define the TD error.

Answer. The target is r plus γ times $\max_{a'} Q(s', a')$ for nonterminal transitions. Delta is target minus $Q(s, a)$, and $Q(s, a)$ increases by α times delta.

Solution 2.b - Q-learning update (1 points)

Question focus. Compute one update for $Q=2$, $r=1$, $\gamma=0.9$, max next $Q=4$, and $\alpha=0.5$.

Answer. The target is $1+0.9*4=4.6$; delta is 2.6; the updated value is $2+0.5*2.6=3.3$.

Solution 2.c - Q-learning update (1 points)

Question focus. Explain and test terminal handling.

Answer. For `done=True` the target is `r` only; no next-state maximum is used. A test gives a very large next `Q` value and confirms it does not change the terminal update.

Solution 3.a - Training protocol (1 points)

Question focus. Describe reset, interaction, update, and episode termination order.

Answer. Reset returns the initial state; each step selects, executes, records reward and done, updates once, and advances state. The loop ends on done or a maximum-step truncation.

Solution 3.b - Training protocol (1 points)

Question focus. Log return, episode length, epsilon, and moving-average return.

Answer. Metrics use consistent windows and distinguish truncation from environment termination. Seeds and hyperparameters are recorded with the run.

Solution 4.a - Evaluation and tests (1 points)

Question focus. Evaluate the greedy policy without learning or exploration.

Answer. Set epsilon to zero, freeze Q values, run independent seeded episodes, and report mean return, success rate, and confidence or variability.

Solution 4.b - Evaluation and tests (1 points)

Question focus. Compare greedy actions or returns with a value-iteration oracle.

Answer. Map both policies to the same MDP semantics. Count exact or tie-equivalent action agreement and explain differences caused by incomplete visitation or stochastic estimates.

Solution 4.c - Evaluation and tests (1 points)

Question focus. Run one controlled hyperparameter ablation and cover sparse-reward and unseen-state cases.

Answer. Vary only alpha, epsilon schedule, or gamma. Report curves and explain bias/noise effects. Unseen states start from a documented default; sparse rewards do not cause crashes or false convergence claims.

Edge-case reasoning and expected behavior

- ▶ Treat it as terminal or raise a domain-contract error.
- ▶ Use target equal to the immediate reward.
- ▶ Choose deterministically in evaluation and uniformly only in exploration.
- ▶ Use the declared initialization value for every legal action.
- ▶ Record truncation separately and stop without a synthetic terminal bootstrap.

Grading rubric

Category	Points	Full-credit evidence
Agent and exploration contract	2	Separation, legal actions, exploration probabilities, and ties are correct.
Q-learning update	3	TD target, terminal mask, and in-place assignment are correct.
Training protocol	2	Episodes terminate safely and training/evaluation are distinct.
Evaluation and tests	3	Metrics are unbiased, controlled, and interpreted correctly.

Reflection exemplar

Answer. A strong response verifies shared MDP semantics first, then uses visitation counts, TD error, or repeated-run evidence to attribute the remaining gap to exploration, finite data, or hyperparameters.

Grading guidance. Award credit for a specific claim supported by technical evidence from the submitted work.